Loan Default Prediction

An interpretable credit-scoring model to approve home equity loans

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

Shubham Garg | November 2026

Key takeaways

1 in 5

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

62%

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is present. Unverified income is the strongest warning sign.

93%

of future defaulters among unseen applicants. The proposed Random Forest model catches them. 2 of every 3 applicants it flags really would have defaulted.

~$9M

indicative annual benefit on 6,000 applications, from roughly $10.8M of default-related cost down to about $1.8M.

The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit. Defaults consume it.
- Today each decision is manual: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act every rejection needs a specific, defensible reason, so the model must be interpretable.

Objective: predict which applicants are likely to default. Tell the bank which attributes to weigh when approving a loan.

The data: Home Equity dataset (HMEQ)

5,960
recent loans

12
attributes known at application

19.9%
defaulted or seriously delinquent

44%
of applications have a blank field

Approach: from raw applications to a decision the bank can defend

1

Explore

Univariate, bivariate and correlation analysis. Default rate by band. Pattern of missing values

2

Prepare

Reserve a stratified 30% test set first. Flag every missing field. Impute median/mode learned on training rows. Cap outliers for the linear model only. One-hot encode

3

Model

Logistic Regression (linear benchmark), Decision Tree, Bagging, Random Forest. Class weights. Cross-validated grid search

4

Measure

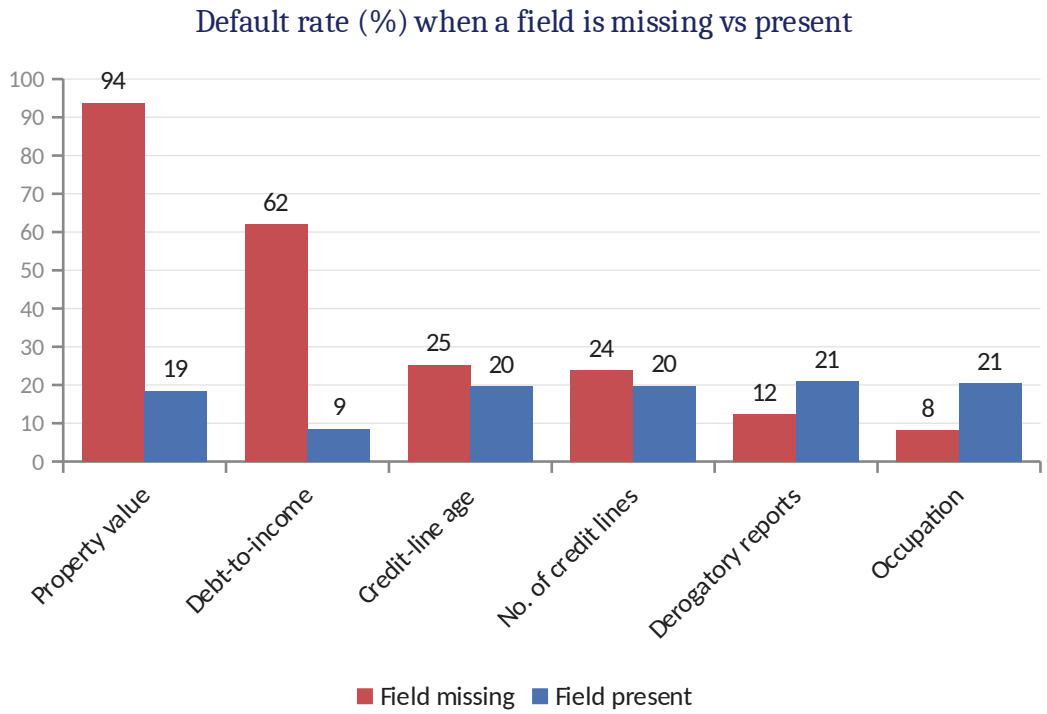
Recall on defaulters as the primary metric. Precision, F1, ROC-AUC as guard-rails. Final model and threshold chosen on out-of-fold expected cost

5

Recommend

Model specification, business rules, implementation plan, benefits, risks

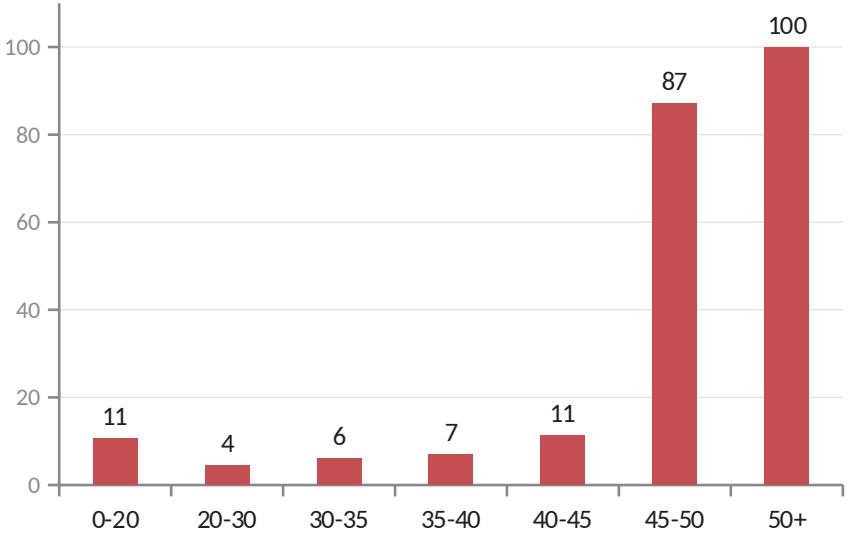
Insight 1: information the applicant did not provide is the strongest warning sign



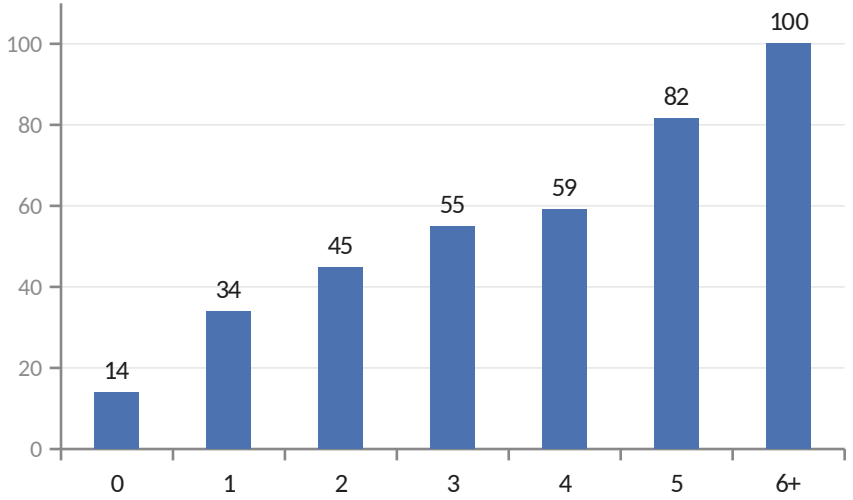
- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data).
- No property valuation: 94% default (112 applicants).
- The overall default rate is 20%, so these gaps multiply risk 3 to 5 times.
- Interpretation: the bank could not check the income or collateral, or the applicant chose not to supply it.
- Action: every imputed field gets a 'was missing' flag in the model. Verification should become mandatory in the process.

Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)



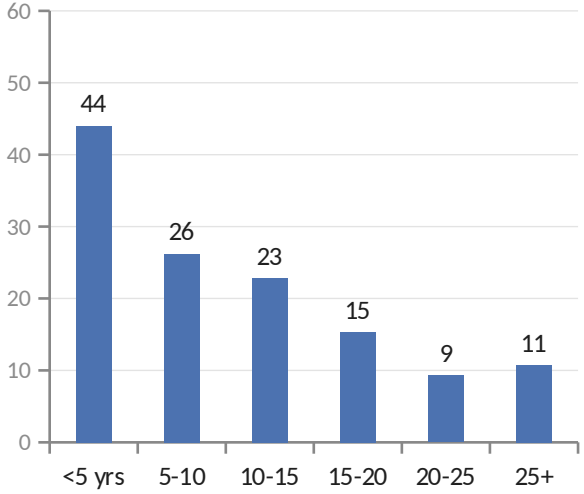
Default rate (%) by number of delinquent credit lines



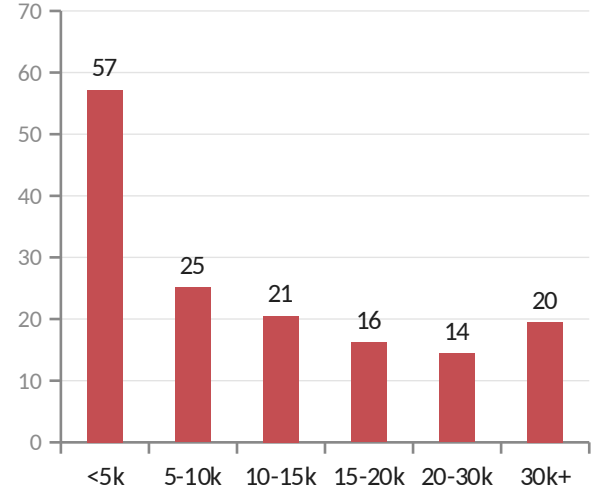
Between a 20% and a 40% ratio only 4-7% default. Above 45% almost everyone does. Each delinquent line or derogatory report roughly doubles the risk (0 reports: 17%, 1: 39%, 2: 51%, 3+: about 75%). Both are plain-language rules to give an applicant.

Insight 3: long credit histories protect, small loans do not, occupation matters less

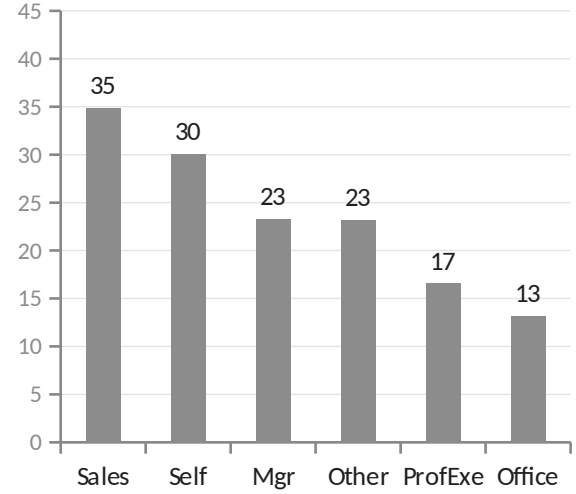
By age of oldest credit line



By loan amount ($)



By occupation



Default rate (%) by group. Applicants with under five years of credit history default four times as often as those with twenty-plus years. Loans under $5,000 default 57% of the time. Occupation shifts risk moderately. On its own it adds little at the usual threshold. But removing it from the final model raises the expected cost of credit errors by about 10%. So it stays, subject to the fair-lending review.

Measure of success: the two mistakes do not cost the same

Missed defaulter

The model says 'will repay'. The customer defaults.

The bank loses most of the principal plus collection cost: about $9,000 on an average $16,900 defaulted loan.

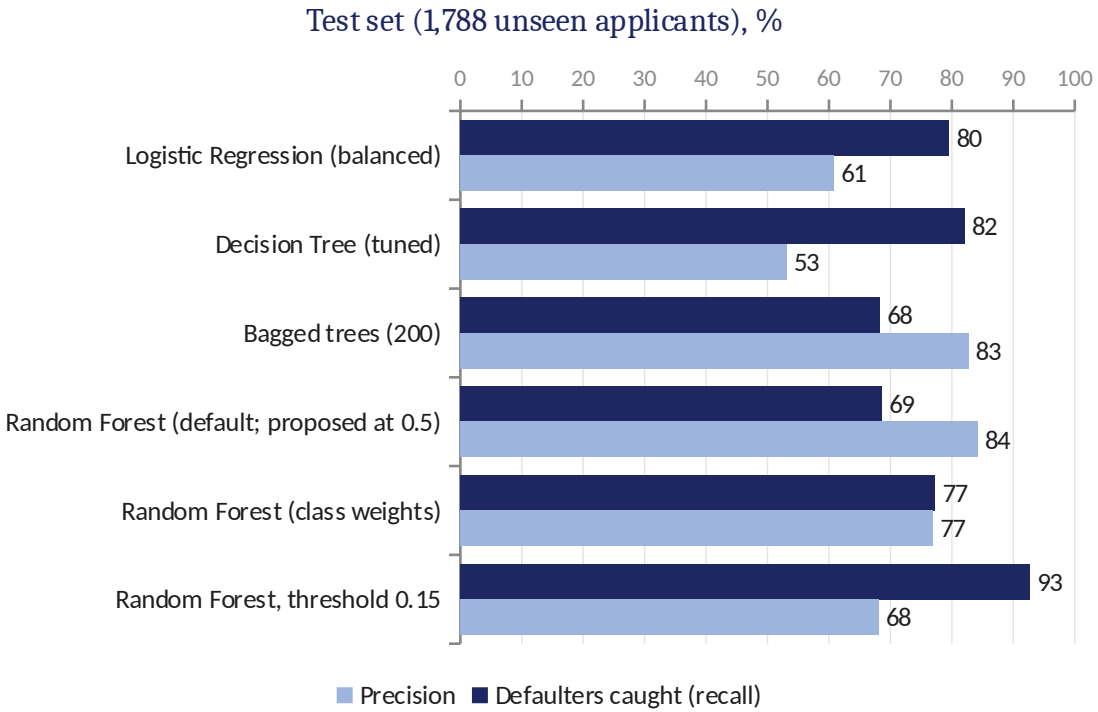
Wrongly declined

The model says 'will default'. The customer would have repaid.

The bank forgoes the interest margin on one good applicant: about $2,000.

- A missed defaulter costs roughly 4.5 times a wrongly declined customer. So the primary metric is Recall: the share of future defaulters the model catches.
- Guard-rails: Precision (how many flagged applicants really default) and F1, so the model does not simply decline everyone. ROC-AUC measures how well the model ranks applicants by risk, which matters when the threshold moves.
- We tune individual models by cross-validated grid search on recall. We choose the final model and its threshold on out-of-fold expected cost, training rows only. We score the 30% test set once.

Model comparison: the Random Forest gives the best balance



- Linear vs non-linear: logistic regression reaches 80% recall but only 61% precision. The data is made of steps, which trees capture and a straight line cannot.
- Single tree vs ensembles: the tuned tree catches 82% but declines almost one good customer per defaulter caught (53% precision). Bagging 200 trees lifts precision to 83% and AUC to 0.95 at 68% recall.
- The class-weighted forest: 77% recall at 77% precision, ROC-AUC 0.96, among the best at ranking applicants by risk.
- Final choice on the expected cost of mistakes (training rows only), by a rule fixed beforehand: lowest mean cost, simplest if within noise. Three forests tied. The plain forest at 0.15 catches 93% at 68% precision on test.

Proposed solution: a Random Forest scorer explained by a surrogate decision tree

Model specification

- Scoring engine: Random Forest, 200 fully grown trees, no class weights: the decision threshold handles the imbalance.
- Inputs: all 27 (10 financial attributes, 11 missing-value flags, occupation and loan purpose). Fallback without occupation: 21 inputs, about 10% more expected cost.
- We chose decision threshold 0.15 together with the configuration to minimise expected cost on repeated out-of-fold predictions of the training rows. It holds for cost ratios 3:1 to 6:1 and across train/test splits. Approve / refer / decline bands (about 69% / 7% / 24% of applicants) keep an underwriter in the loop.
- Explanation layer: a depth-4 surrogate tree fitted to the forest's out-of-fold decisions reproduces 93% of them. Its rules (ratio missing, ratio above 44%, a delinquent line together with a derogatory report) supply the ECOA adverse-action reasons.
- We scored the reserved test set only once, after all choices.

93%
of defaulters caught

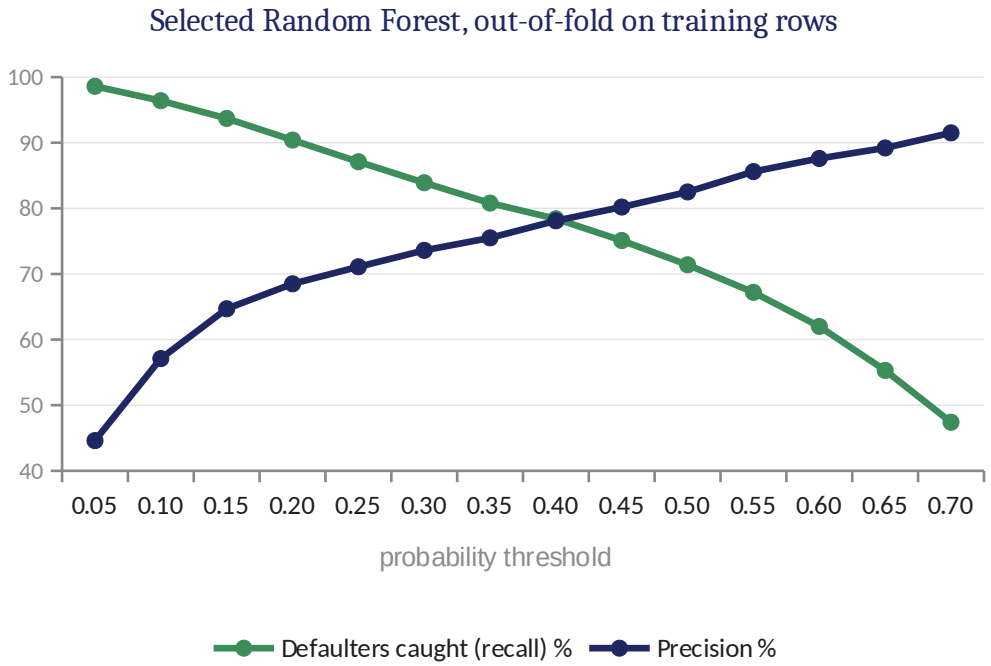
68%
of flagged applicants truly default

0.96
ROC-AUC: risk ranking quality

90%
overall accuracy

Per 1,000 applicants: 185 of 200 defaulters flagged in advance, 15 missed, 87 good customers referred or declined. 27% of applicants flagged, 73% approved.

Why a 0.15 threshold: it is a business dial, not a constant



At 0.50 (default)

- 63 missed defaulters
- 26 good customers declined
- $615k indicative cost

At 0.15 (proposed)

- 15 missed defaulters
- 87 good customers declined
- $304k indicative cost

Curves: out-of-fold predictions on the training rows, where we chose the threshold. Cards: test set, per 1,000 applicants. Cost assumes $9,000 per missed defaulter and $2,000 per wrongly declined customer. The optimum stays at 0.15 for cost ratios from 3:1 to 6:1, on a finer grid and on other train/test splits. It sits just below the textbook break-even (0.18) because the scores understate risk.

Business impact: an indicative $9 million a year

Per year, 6,000 applications	Current process	With the proposed model
Loans that default	about 1,200 (20%)	about 90 missed
Default-related losses	about $10.8M	about $0.8M
Good customers declined or referred	none	about 520, margin forgone about $1.0M
Total cost of credit errors	**about $10.8M**	**about $1.8M**

~$9M

indicative gross annual benefit. About $8.9M after running costs, $8.5M in year one

- Assumptions: 6,000 applications a year. Defaulted loans average $16,900 with about half recovered ($9,000 per missed defaulter). Forgone margin about $2,000 per declined good customer. Production performance equal to test performance. Measured as an extra screen on the loans the bank approves today, before mandatory income verification.
- Costs: roughly $300-400k one-off (pipeline, integration, validation, fairness review, training) and about $100k a year to run and monitor. Repays its cost within year one even if only a quarter of the benefit is realised.
- Non-financial: faster decisions, consistent treatment, auditable reasons for every decline, verified income and valuation on every application.

Recommendations and implementation roadmap

1 Months 0-2: prepare

- Make income and valuation verification mandatory fields
- Build the scoring pipeline and reason codes
- Check the threshold with real loss and margin figures
- Review fairness and model risk

2 Months 2-4: shadow and pilot

- Score every application alongside the manual decision
- Re-train once verified ratios arrive. Tune the approve / refer / decline bands
- Train underwriters on reason codes and referrals

3 Month 4 onwards: scale

- Start automated approve / decline for clear cases
- Monitor monthly: recall, precision, approval rate, missing fields
- Yearly re-training. Threshold review with finance

Policy rules to adopt now, independent of the model

- Decline or escalate above a 45% debt-to-income ratio. Refer to an underwriter with two or more delinquent credit lines or two or more major derogatory reports. Observed default: 45% and 51%+. Hard rules take precedence over the score.
- Treat an application with no verifiable income or property valuation as high risk, never as neutral.

Risks, challenges and further analysis

Risks and mitigations

- Selection bias: the data covers only loans the old process approved. Mitigate with a shadow period on all applications and reject inference.
- The 'ratio missing' signal will change once verification is mandatory. Re-train on the verified population before go-live. On applicants with a recorded ratio the model already catches 83% at 80% precision.
- Economic drift: monitor monthly, re-train yearly, review the threshold.
- Fair lending: the 0.15 threshold wrongly flags about 11% of creditworthy applicants (3% at 0.5). Test for disparate impact before go-live, yearly and at each threshold change. A fallback without occupation is ready (about 10% more cost).
- Explainability and automation bias: tree-based reason codes, a referral band, documented overrides.

Further analysis

- Reject inference to cover applicants the old process declined.
- Engineered features: loan-to-value and payment-to-income ratios.
- Gradient boosting and probability calibration, so a score of 0.15 means exactly a 15% chance of default.
- Time-based validation once application dates are available.
- Risk-based pricing for the referral band. Early-warning scoring on existing loans.
- SHAP-style reason codes per applicant in applicant-friendly language.

Thank you

Questions and discussion

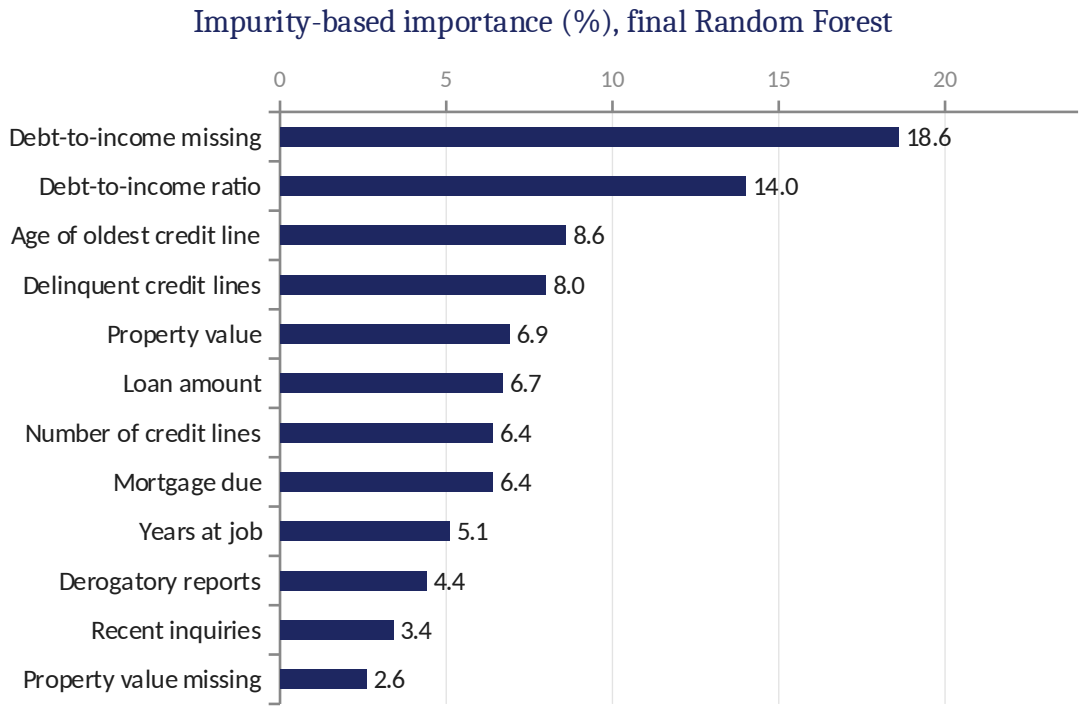
Catch 93% of defaulters. Explain every decision. Save an indicative $9M a year.

Appendix A: all models on the held-out test set

Model	Recall	Precision	F1	Accuracy	ROC-AUC	Train recall
Logistic Regression (plain)	0.566	0.774	0.654	0.880	0.907	0.635
Logistic Regression (balanced, C tuned)	0.796	0.608	0.689	0.857	0.906	0.821
Decision Tree (default)	0.686	0.706	0.696	0.880	0.807	1.000
Decision Tree (class weights)	0.616	0.719	0.664	0.875	0.778	1.000
Decision Tree (tuned: balanced, min 20 per leaf)	0.821	0.531	0.645	0.819	0.877	0.921
Bagging (200 fully grown trees)	0.683	0.827	0.748	0.908	0.945	1.000
Bagging (class-weighted trees)	0.597	0.810	0.687	0.891	0.946	1.000
Random Forest (default, 200 trees)	0.686	0.842	0.756	0.912	0.963	1.000
Random Forest (class weights 0.2/0.8)	0.773	0.769	0.771	0.908	0.959	1.000
Random Forest (tuned for recall)	0.790	0.656	0.717	0.875	0.932	0.918
Random Forest (tuned, 12 features)	0.784	0.660	0.717	0.876	0.929	0.916
Proposed: RF unweighted, 27 inputs, threshold 0.15	**0.927**	**0.681**	**0.785**	**0.899**	**0.963**	-

Test set of 1,788 applicants at the 0.5 threshold except the proposed model. Train recall of 1.000 marks un-pruned models that memorised the training rows. We chose the proposed model (the default forest with its threshold moved) and its threshold on repeated out-of-fold expected cost (training rows only). We fixed the rule beforehand: lowest mean cost, simplest set-up among those within one standard error (about $10,500). Mean out-of-fold ROC-AUC 0.961 against 0.963 on test.

Appendix B: what the Random Forest relies on



- The debt-to-income ratio and its missing flag carry a third of the final forest's decision power. With delinquent lines and credit-history length, about half.
- Permutation importance (drop in ranking quality when we shuffle a column) ranks the same four at the top, delinquent credit lines first.
- Loan, property and mortgage amounts matter in combination, little on their own.
- Occupation, loan purpose and minor flags matter little individually. Yet removing them raised out-of-fold expected cost by about a fifth, and removing occupation alone by about 10%. So they stay subject to the fair-lending review.
- Logistic-regression coefficients, tree rules and the surrogate tree all point to the same drivers.

Appendix C: the explanation layer, rules of the surrogate tree that follows the forest

1	**Is the debt-to-income ratio missing?**	Yes: flag, whatever the credit history (62% of these applicants defaulted). Check income before approving.
2	**If recorded, is the ratio above about 44%?**	Yes: flag (87-100% of applicants above 45% defaulted).
3	**Ratio at or below 44%: a delinquent credit line together with a major derogatory report?**	Yes: flag. Otherwise clear. Applicants with no delinquent line default only about 4-5% of the time.
4	**How faithful are these rules?**	They reproduce 93% of the forest's decisions on unseen applicants (73% for never flagging, kappa 0.82). The forest's own top risk drivers explain the remaining 7%.
5	**What the rules do not use**	Loan size, credit-history length, occupation and loan purpose do not change the outcome in the surrogate. They matter only in combination inside the forest, which supplies the per-applicant reason codes.

Surrogate: depth-4 tree, minimum 20 applicants per leaf, fitted to the forest's out-of-fold flag/clear decisions at threshold 0.15 on the training rows. It agrees with the forest on 93% of test decisions (73% for a rule that never flags). The stand-alone tuned tree (test recall 82%, precision 53%) found the same drivers.

Appendix D: data treatment and pre-processing

Step	What we did	Why
Duplicates	None found. No rows removed	5,960 loans retained, including 2 with only the loan amount recorded
Missing values	A 0/1 'was missing' flag per column, then median (numeric) or mode (categorical) imputation learned on the training rows	Missingness of debt-to-income and property value is the strongest default signal. Removing rows would lose 44% of the data
Outliers	IQR capping for the linear model only. DEROG and DELINQ capped at the 99th percentile. Tree models use raw values	Extreme values pull linear models (debt-to-income 203%, credit history 97 years). Trees split on rank order and keep the 'above 50% always defaults' signal
Encoding and scaling	One-hot encoding of REASON and JOB (drop first). Standard scaling of the continuous inputs for the linear model only, flags left as 0/1	Comparable coefficients, odds ratios for flags read as the effect of the flag
Split	70% train / 30% test, stratified on the target, fixed seed, reserved before we fitted any statistic. Identical applicants in both data versions	No leakage from test rows into fences, medians or scaling. Results comparable across models
Imbalance	Class weights (balanced or 0.2/0.8) inside the individual models, then the decision threshold for the final model. No resampling	Keeps each real applicant. Class weights lifted recall by 23 points for logistic regression and 9 for the forest at the 0.5 threshold. They made no measurable difference once we chose the threshold on cost, so the final forest is unweighted
Model inputs	27 columns: the 10 attributes, 11 missing flags and 6 one-hot indicators (REASON HomeImp, JOB Office/Other/ProfExe/Sales/Self)	The notebook's final-pipeline cell names every input to the final model, with the imputation values and model settings